Job Transformation, Specialization, and the Labor Market Effects of AI

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October 15, 2026

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- Our focus: **automation** shifts the task content of jobs $\sim$ **job transformation**

Job transformation: LLMs in the legal profession and beyond

Industrial Revolution

The AI Shift Professional services + Add to myFT

The AI Shift: If AI is coming for junior lawyers' jobs, why does their pay keep going up?

Automation may be freeing up their time to do more valuable work

Junior associates [...] will historically have been doing a lot of document review work [...] A lot of that can be automated [...] those associates can be used more efficiently on other things [...] [like] interacting with clients

Financial Times, Dec 18, 2025

The New York Times Magazine

[They] are still software developers, but like most of their peers now, they only rarely write code. Instead, they spend their days talking to the A.I., describing in plain English what they want from it [...]

The New York Times Magazine, Mar 12, 2026

The executive assistant role isn't dying. It's getting promoted



Fortune, Jun 22, 2026

AI scribes save 15,000 hours—and restore the human side of medicine

After 2.5 million uses in one year, The Permanente Medical Group's ambient AI scribes ease documentation burden, reduce burnout, and improve communication.

By Benji Feldheim, Contributing News Writer

Jun 11, 2025 7 Min Read

AMA News Wire, Jun 12, 2025

FEATURES

AI isn't replacing radiologists

Radiology combines digital images, clear benchmarks, and repeatable tasks. But demand for human radiologists is at an all-time high.



WORKS IN PROGRESS AND DEENA MOUSA

SEP 25, 2025

Works in Progress, Sep 25, 2025

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- Our focus: **automation** shifts the task content of jobs $\sim$ **job transformation**
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- **This paper:** a framework to quantify the wage effects of AI-induced job transformation for heterogeneously skilled workers

A framework to quantify the labor market effects of AI

- **Theory**: formalize **job transformation wage effects** in a task-based GE model

- **Measurement:** estimate **skill distribution** via MLE using model structure

- **Application:** project worker-level wage effects due to **AI** (“if-then” analysis)

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  - scenario: LLMs automate information-processing tasks

Who wins and who loses? Three key takeaways.

1 **Occupational exposure** · What are the wage effects of being in an AI-exposed job?

On average, moderate AI exposure raises wages, high exposure lowers them.

2 **Worker reallocation** · Can workers adjust by moving to other jobs?

Some workers face large losses, with reallocation offering limited insurance.

3 **Skills** · Which skills will AI make more or less valuable?

AI raises the value of social skills and lowers that of some analytical skills.

Contributions to the literature: brief overview

▸ Literature in detail

- **Task-based theory** *[e.g., Acemoglu-Autor, 2011; Acemoglu-Restrepo, 2018; Acemoglu-Restrepo, 2022]*

  $\Rightarrow$ empirically operationalize (occupations as task bundles + occupational choice), enabling estimation and quantitative scenario analysis

- **Multi-dimensional skills** *[e.g., Lindenlaub, 2017; Lise-PostelVinay, 2021; Baley et al., 2022]*

  $\Rightarrow$ estimate granular skills, bridging the gap to task-based technical change

- **Labor market effects of AI** *[e.g., Hampole et al., 2025; Autor-Thompson, 2025; Korinek et al., 2026]*

  $\Rightarrow$ highlight & quantify importance of job-transformation mechanism

Theory

Model environment: task-based production meets Roy

CES | Equilibrium definition

Technology

Final good: CES (σ) aggregator over occupational output

Occupation: $o \in \mathcal{O}$ **bundles** tasks $\mathcal{T}$ with weights $\{\alpha_{o,\tau}\}_{\tau \in \mathcal{T}}$

$$Y_{i,o,t} = \prod_{\tau \in \mathcal{T}} X_{i,\tau,t}^{\alpha_{o,\tau}}$$

where $X_{i,\tau,t} = X_{i,\tau,t}^{\text{machines}} + X_{i,\tau,t}^{\text{labor}}$

Tasks: Assigned to labor ($\rightarrow \mathcal{T}_l$) or machines ($\rightarrow \mathcal{T}_m$)

Firm/job: Hires 1 worker (i), chooses machine quantity $M_{i,\tau,t}$ $\rightarrow X_{i,\tau,t}^{\text{machines}} = \exp(z_\tau) M_{i,\tau,t}$

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Workers

Skills: Task-specific, time-invariant

$$s_i \equiv \{s_{i,\tau}\}_{\tau \in \mathcal{T}_l} \sim \mathcal{N}(\bar{s}, \Sigma_s)$$

$$\rightarrow X_{i,\tau,t}^{\text{labor}} = \exp(s_{i,\tau}) \ell_{i,\tau,t}$$

Occupational choice:

$$\hat{o}_{i,t} = \text{argmax}_o w_{i,o,t} + u_{i,o,t},$$

$$u_{i,\cdot,t} \sim \text{Gumbel}(0, \nu)$$

Time: Inelastic unit supply

$$\sum_{\tau \in \mathcal{T}_l} \ell_{i,\tau,t} = 1$$

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Markets

Labor: Competitive

Capital: Infinitely elastic supply, machines can be rented at rate r

Occupational output: Competitive

Final good: Competitively traded numeraire

Optimal time allocation is proportional to weight matrix A

Firm problem · Capital FOC

Firm's profit maximization problem yields:

$$\ell_{i,\tau,t} = \frac{\alpha_{o,\tau}}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}} = \boxed{\frac{\alpha_{o,\tau}}{\mathsf{LS}_o}}$$

The **occupational task-weight matrix** A summarizes relative weights attached to tasks $\tau \in \mathcal{T}_l$ across occupations $o \in \mathcal{O}$:

$$A = \begin{pmatrix} \frac{\alpha_{1,1}}{LS_1} & \frac{\alpha_{1,2}}{LS_1} & \cdots & \frac{\alpha_{1,n_{\text{skill}}}}{LS_1} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\alpha_{n_{\text{occ}},1}}{LS_{n_{\text{occ}}}} & \frac{\alpha_{n_{\text{occ}},2}}{LS_{n_{\text{occ}}}} & \cdots & \frac{\alpha_{n_{\text{occ}},n_{\text{skill}}}}{LS_{n_{\text{occ}}}} \end{pmatrix} \in \mathbb{R}^{|\mathcal{O}| \times |\mathcal{T}_l|}$$

where $\mathsf{LS}_o = \sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}$ denotes the labor share in occupation o.

Model yields a tractable and intuitive log-linear wage equation

▸ Intercept term

$$w_{i,\cdot,t} = \mu + A s_i + \varepsilon_{i,t}$$

$$\underbrace{w_{i,o,t}}_{\text{log wage of } i \text{ if choose } o \text{ in } t} = \underbrace{\mu_o}_{\text{occ.-specific intercept (incl. prices)}} + \overbrace{\sum_{\mathcal{T}_l} A_{o,\tau} \cdot s_{i,\tau}}^{\text{weighted skills}} + \underbrace{\varepsilon_{i,t}}_{\text{idiosyncratic productivity shock (not crucial)}}$$

$$= \mu_o + \underbrace{\frac{1}{|\mathcal{T}_l|} \sum_{\mathcal{T}_l} s_{i,\tau}}_{\text{scalar absolute advantage } \perp o} + |\mathcal{T}_l| \cdot \underbrace{\text{Cov}\left(A_{o,\cdot},\ s_{i,\cdot}\right)}_{\text{comparative advantage in } o} + \varepsilon_{i,t}$$

Automation leads to job transformation *given* task bundling

- **Automation**: rise in machine productivity $z_{\tau^\star}$ making it optimal to reassign $\tau^\star$

$$\mathcal{T}_l' = \mathcal{T}_l \backslash \tau^\star \qquad \mathcal{T}_m' = \mathcal{T}_m \cup \tau^\star$$

- **Job transformation** = change in A_o: weight on $\tau^\star \downarrow$ & weight on other tasks $\uparrow$ proportional to their occupation-specific weight
  - "**partial automation**": only a fraction $\zeta_{\tau^\star} \in [0, 1]$ is automated

Details

	$\tau^\star$	Task 2	Task 3	Task 4	Task 5
A_o **(Pre)**	0.2	0	0.4	0.1	0.3
A_o **(Post)**	0	0	0.5	0.125	0.375

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- Job transformation meaningful ($A_o' - A_o \neq 0$) if an occ. features **task bundling**:

$$|\{\tau \in \mathcal{T}_l : \alpha_{o,\tau} > 0\}| > 1$$

Wages change due to occupation-level *and* job-transformation effects

$$\underbrace{\mathbb{E}\left[w_{i,o,t+1} - w_{i,o,t}\right]}_{\Delta \text{ potential log wage for } i \text{ in } o} = \Delta\mu_o + \underbrace{(A'_o - A_o)}_{\Delta \text{ task weights in } o} \cdot \underbrace{s_i}_{\text{individual skill vector}}$$

1 Occupation-level effects:

$$\Delta\mu_o = \underbrace{\frac{\alpha_{o,\tau^\star}}{LS_o - \alpha_{o,\tau^\star}}\left(z_{\tau^\star} - \log r + \mu_o\right)}_{\text{occupation-level automation effect}} + \left(\underbrace{\frac{\log P'_o - \log P_o}{LS_o - \alpha_{o,\tau^\star}}}_{\text{GE effect}}\right)$$

2 Job transformation effects:

$$(A'_o - A_o)\cdot s_i = \underbrace{A_{o,\tau^\star}}_{\text{occupational exposure}} \times \left(\sum_{\mathcal{T}_l \setminus \tau^\star} \underbrace{A'_{o,\tau} s_{i,\tau} - s_{i,\tau^\star}}_{i\text{'s relative specialization}}\right)$$

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Exposure signals the potential for change; skills determine its direction.

Measurement

Measurement

Validation: task weights | Task → cluster examples | A_{pre}

→ We parametrize the model at same 'resolution' as task-exposure measures

- **Step 1: map model tasks & occupations to data, *construct* A**
  - tasks: NLP tools to cluster ∼19,000 O*NET task statements → 38 task clusters Details
  - occupations: 3-digit, SOC-2000
  - occ. task weights (A): identified off of LLM-generated time diaries given $\ell_{i,\tau,t} = \frac{\alpha_{o,\tau}}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}} := A_{o,\tau}$
  - robustness: aggregated O*NET weights
  - time-varying (pre-2000, post-2000)

- **Step 2: *estimate* skill distribution** $(\bar{s}, \Sigma_s)$ Details
  - data: A + NLSY '79 panel Details
  - identifying variation: wage variances within occupations (stayers), wage covariances across occupations (switchers), occupational choices
  - implementation: maximum likelihood estimation (MLE)

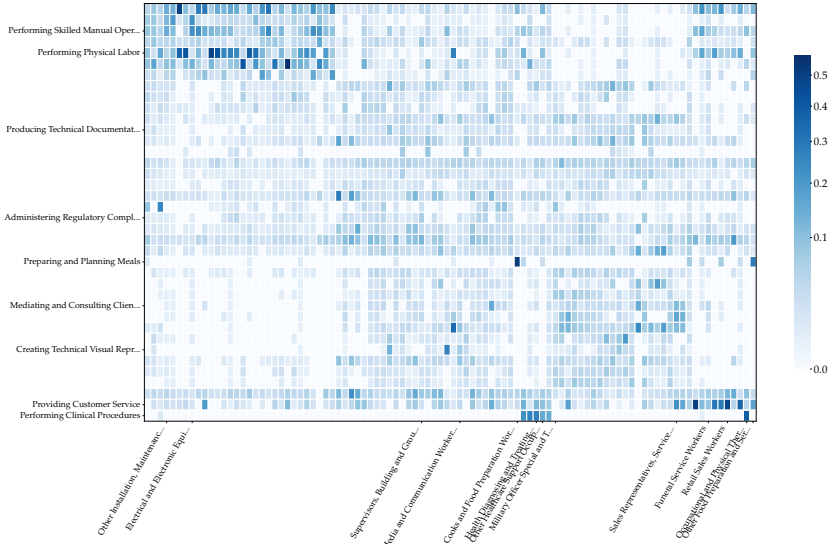
Measurement

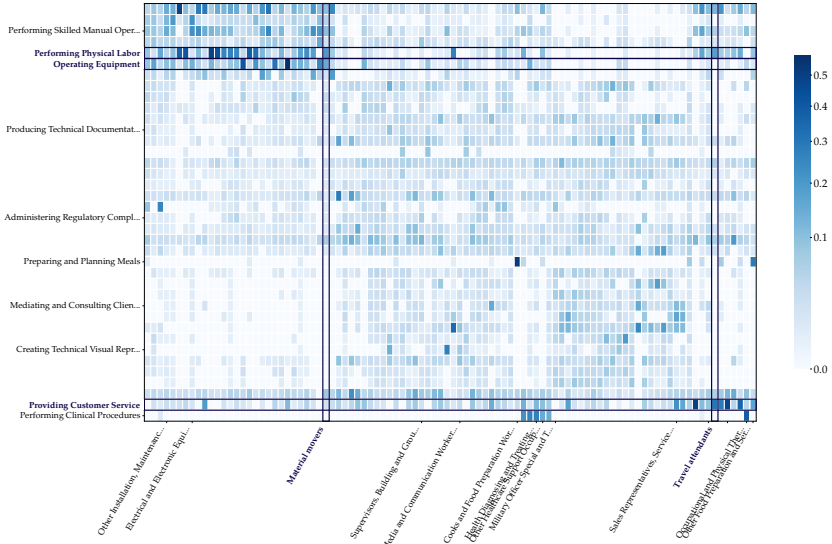
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Performing Physical Labor
Producing Technical Documentat...
Administering Regulatory Compl...
Preparing and Planning Meals
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Performing Clinical Procedures
Other Installation, Maintenanc...
Electrical and Electronic Equi...
Supervisors, Building and Grou...
Media and Communication Worker...
Cooks and Food Preparation Wor...
Health Diagnosing and Treating...
Other Healthcare Practitioner...
Military Officer Special and T...
Sales Representatives, Service...
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Other Food Preparation and Se...
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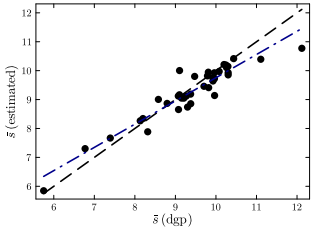
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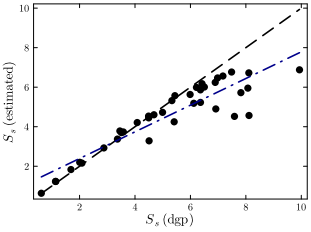
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Validation: Monte-Carlo study

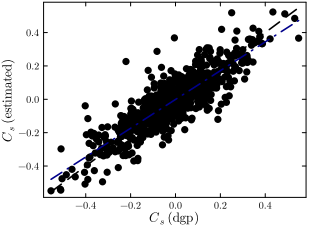
(a) Means



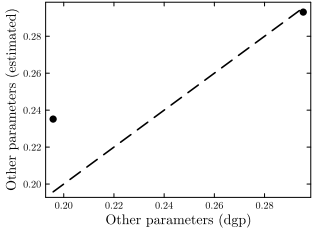
(b) Standard deviation



(c) Correlation



(d) Other parameters



Model properties & validation: overview

Parameter values

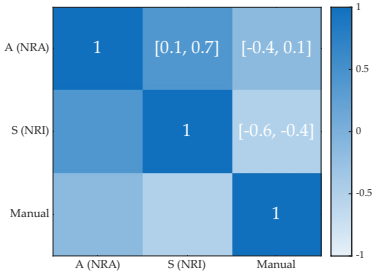
- **Parameter estimates**
  - Skill correlations vs. literature Details
  - Skills along the wage distribution vs. literature Details
- **Steady state**
  - Occ. employment shares Details
  - Wages Details
    - Occ. level averages
    - Between-within occ. variance decomposition
  - Occ. transitions:
    - Staying and switching probabilities Details
    - Switches explained by task distances Details
    - Individual switching frequency shaped by specialization Details
- **Historical: RBTC**
  - Skill returns: ↑ return to social skills *[Deming, 2017]* Details
  - Employment: polarization *[Autor & Dorn, 2013]* Details

Estimated skill correlations match empirical literature

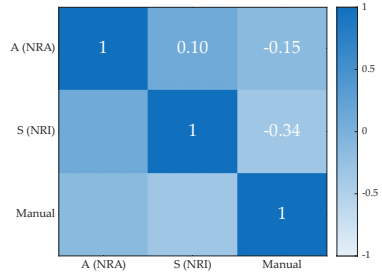
- Literature studies coarser task classification, esp. NRA/NRI/NRM/RC/RM *[e.g., Autor et al., 2003]* – we map our granular tasks to this classification & create skill indices

→ Estimated correlations match empirical ranges for aggregated skills

[e.g.,Deming, 2017; Guvenen et al., 2020; Lise & Postel-Vinay, 2020; Barany & Holzheu, 2025]



(a) Literature summary

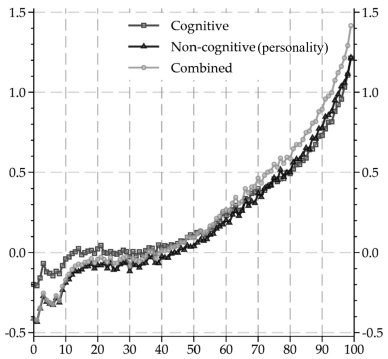


(b) Model

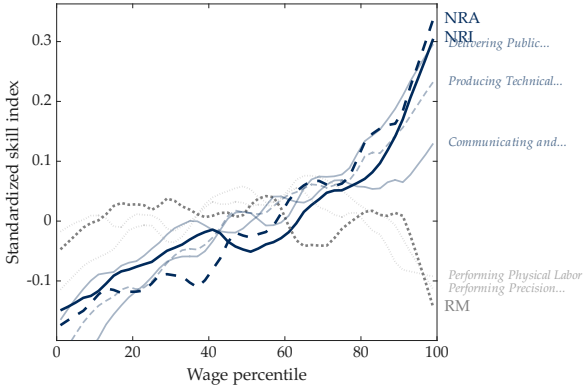
Skill dispersion along the wage distribution in line with data

→ In model & data, analytical and social (but not manual) skills rise along the wage dist.

(a) Data *[Bratsberg et al., 2025]*

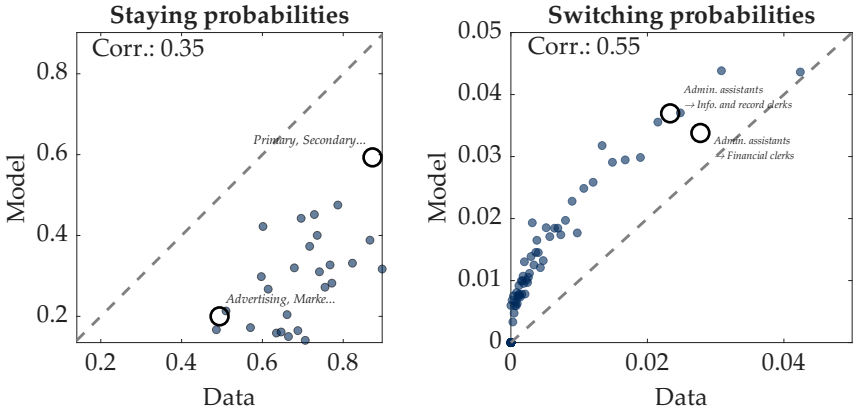


(b) Model



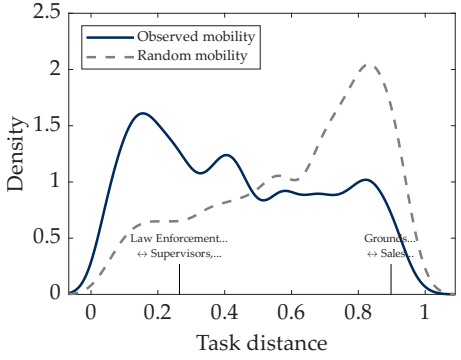
Model endogenously generates plausible occupational transitions

→ Model endogenously generates (some) persistence (we include no exogenous switching costs) & directionally tracks switching patterns

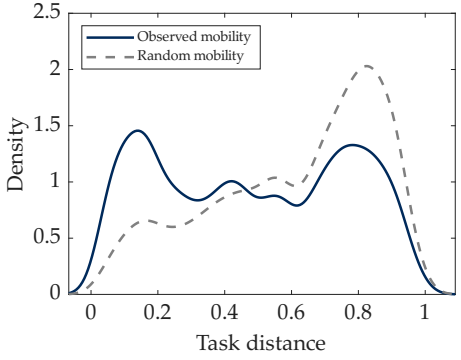


Task requirements explain switching in model and data

→ In model & data, workers move to occupations with similar task requirements—evidence for the relevance of task-specific skills *[Gathmann-Schoenberg, 2010]*



(a) Data



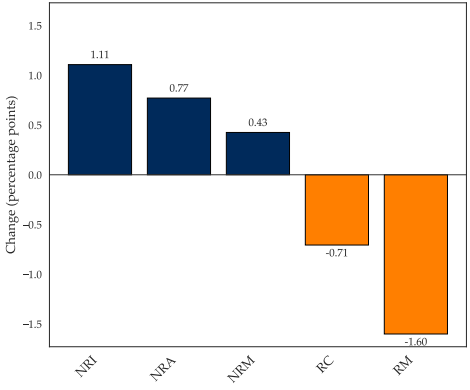
(b) Model

Historical validation: RBTC

- Large empirical literature on "**routine-biased technological change**" (RBTC)
- Compare pre- vs. post-2000 *A*

→ NR (social & analytical) tasks gain in relative weight

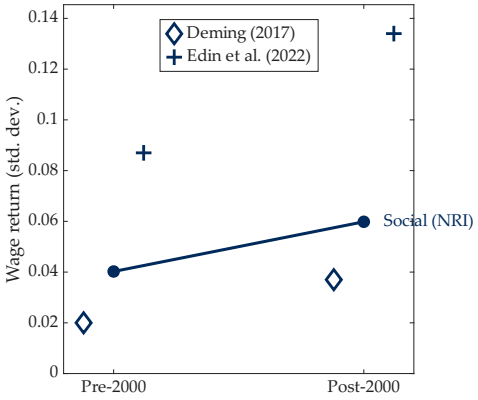
▸ Task trends vs. Atalay et al. (2020)



Historical validation: RBTC drives a rising return to social skills

- Large empirical literature on "**routine-biased technological change**" (RBTC)
- **Model analysis:** compare SS under post- vs. pre-2000 A's

$\rightarrow$ Model reproduces empirically observed $\uparrow$ return to "social" skills *[e.g., Deming, 2017]*



AI

We use the model to project the wage effects due to genAI automation

- **If-then analysis:** If AI automates certain tasks, what are effects for individuals?
  - welfare-relevant *individual*-level effects rather than *occupations*

- **Measurement challenge:** which specific tasks will be/are being automated?
  - want to be able to conduct scenario analysis

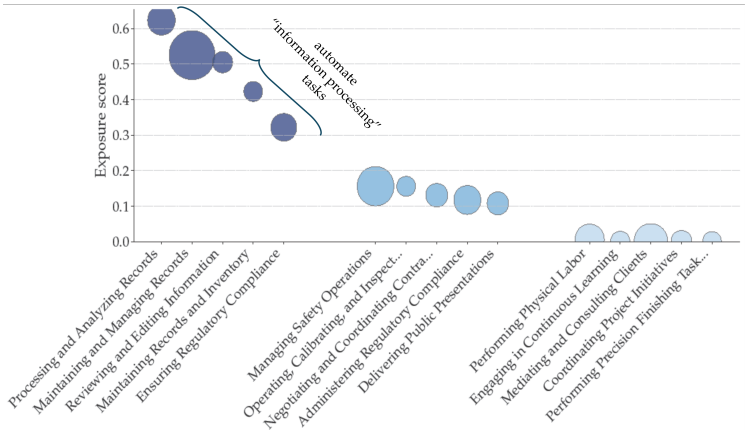
  → **solution:** exploit mapping of model tasks to task exposure measures

- **Flexible framework**
  - maps to many exposure measures *[Webb, 2019; Eloundou et al., 2024; Handa et al., 2024; ...]*
  - can be applied to various technologies: genAI, self-driving vehicles, humanoid robots, ...

- **This paper:** focus on automation due to **LLMs**
  - scenario where $z_{\tau^\star}$ is *just* high enough for a task to be automated in all occ.'s

Scenario: LLMs mostly automate information-processing tasks

→ We suppose a fraction of each task proportional to the exposure score is automated



*Share of detailed O*NET tasks within each task cluster rated as fully automatable by Eloundou et al. (2024).*

We use the model to address 3 key questions

1. **Occupational exposure.** What are the wage effects of being in an AI-exposed job?

2. **Worker reallocation.** Can workers adjust by moving to other jobs?

3. **Skills.** Which skills will AI make more or less valuable?

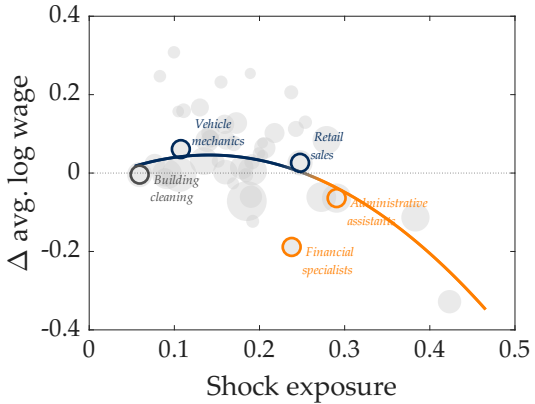
1 **Occupational exposure.** What are the wage effects of being in an AI-exposed job?

Exposure: on average some exposure is good but too much is bad

PE

[+] moderate exposure → AI complements what incumbents are best at
[-] high exposure → AI takes over what incumbents are best at

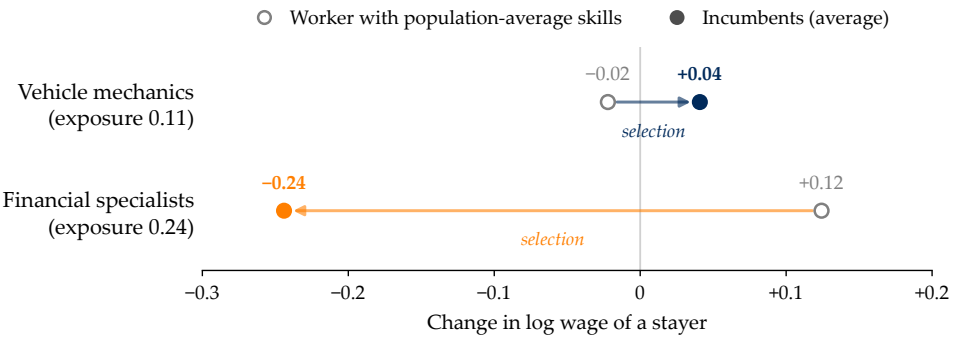
Details



Vertical axis indicates change in individual wage, horizontal axis is the exposure of the origin occupation.

Example: the role of selection

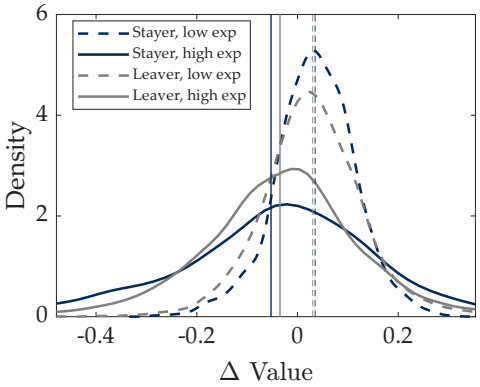
→ Workers in heavily exposed occupations typically lose, as they were selected for the automated tasks, whereas those in moderately exposed occupations are "augmented"



Stayer's log wage change: $\Delta w_{i,o} = \Delta\mu_o + (A'_o - A_o)\, s_i$. *Open circle:* $s_i = \bar{s}$, *the population mean. Filled circle: mean over the occupation's pre-shock workers,* $\bar{s}_o$. *Arrow: selection,* $(A'_o - A_o)(\bar{s}_o - \bar{s})$.

From occupational averages to dispersion across individuals

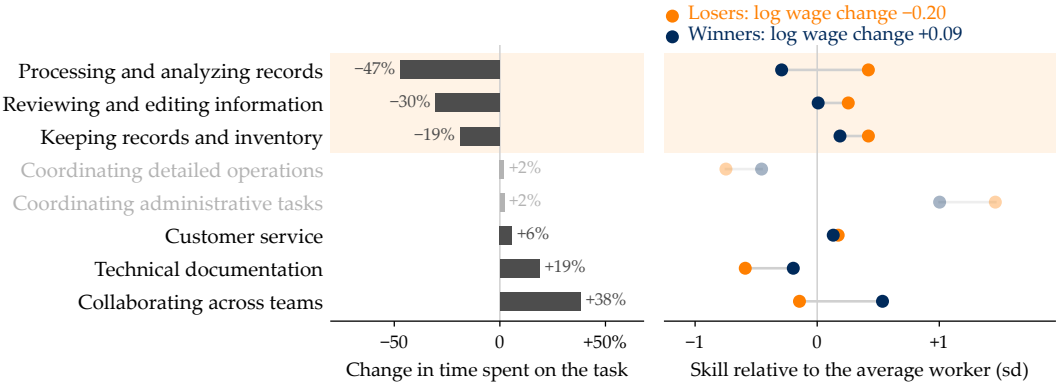
→ Both the biggest individual gains *and* losses are among stayers in high-exposure occ's



"Value" is inclusive value (log-sum) of the worker's occupational wage prospect: $V_{i,t} = \nu \log\left(\sum_{o \in \mathcal{O}} \exp\left(\frac{w_{i,o,t}}{\nu}\right)\right)$.

Example: winners and losers among office job workers

→ Faced with the same shock but possessing different skill profiles, some administrative assistants gain while others lose



Left: change in task weight $A_{o,\tau}$. Right: mean task skills, in population std. dev., of administrative assistants (SOC 43-6). 2 groups: bottom half ("losers") and top half ("winners") of the log wage change as a stayer, $\Delta w_{i,o}$.

1 **On average, moderate AI exposure raises wages, high exposure lowers them.**

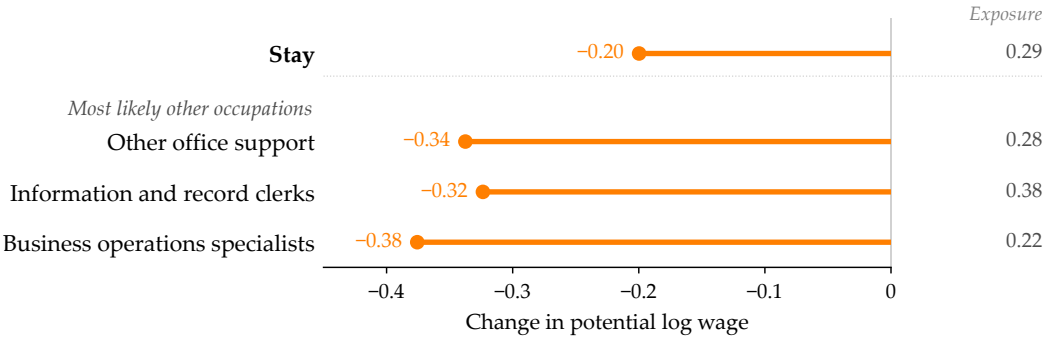
Task-level automation can be worker-level augmentation.

2 **Worker reallocation.** Can workers adjust by moving to other jobs?

Example: “trapped” office workers

▸ Gains for in-switchers

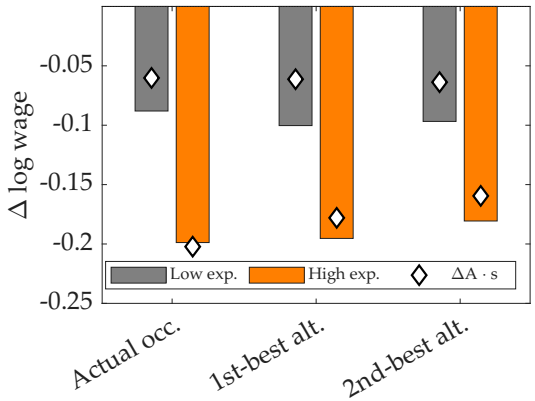
→ Administrative assistants facing a wage drop would usually move to other office jobs – but the AI shock alters skill requirements in *all* these office jobs.



Subsample of stayers in admin. assistant occ. in bottom-half of log wage changes; average change $\Delta w_{i,o}$. Most likely other occupations = highest pre-shock choice probabilities.

The insurance value of reallocation is limited following automation shocks

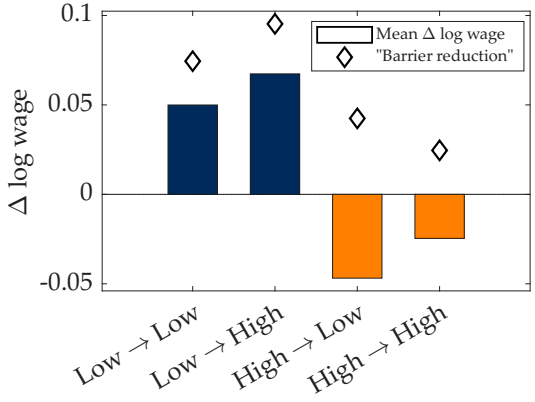
$\rightarrow$ Job transformation leads to correlated $\downarrow$ in potential wages across likely occupation choices, eroding the insurance value of reallocation.



Sub-sample of stayers experiencing negative wage changes.

Switching from low- *into* high-exposure occ.'s yields gains

→ Transformed occ.'s pull in new workers whose skills align better with new task profile



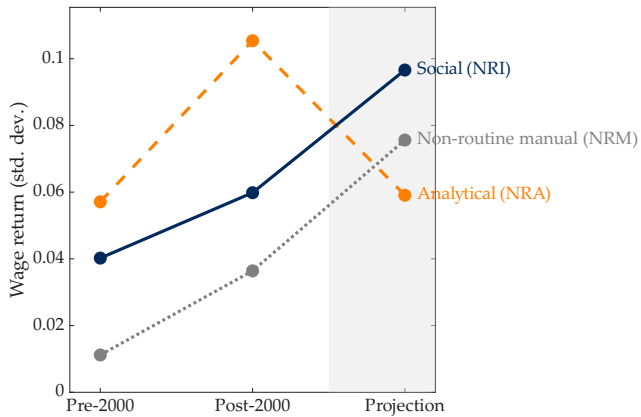
Sub-sample of occupational leavers.

2 **Some workers face large losses, with reallocation offering limited insurance.**

Task automation transforms losers' usual alternative occupations in the same way.

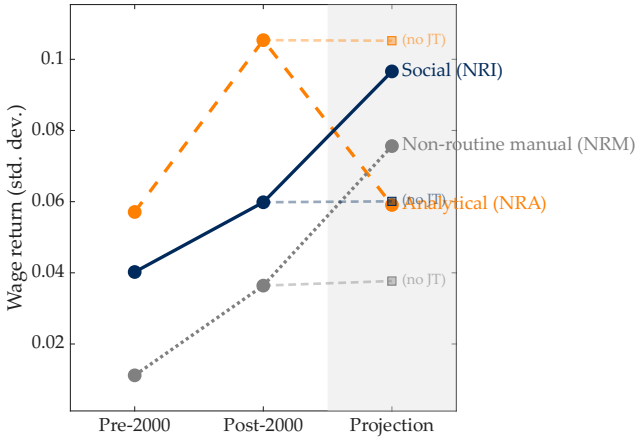
3 **Skills.** Which skills will AI make more or less valuable?

Skills: social & NRM skills gain, analytical skills lose



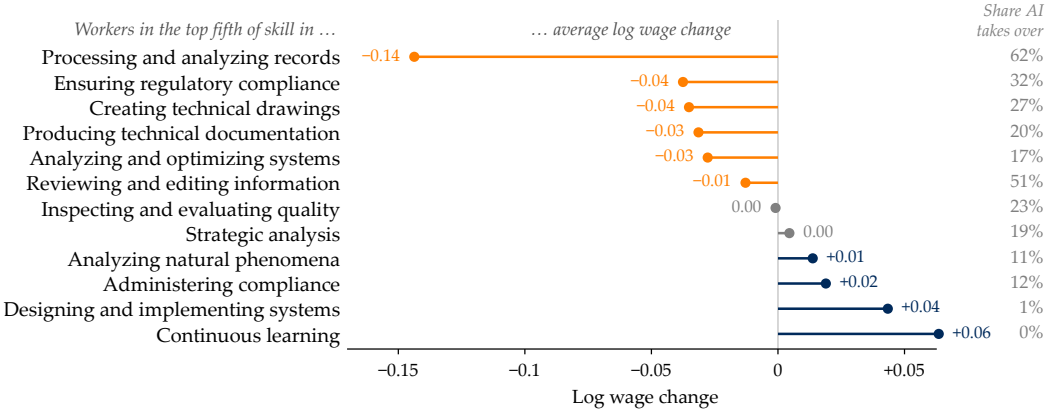
Skills: the shift in returns comes from job transformation

→ In exposed jobs, job transformation shifts weight toward social tasks.



Unpacking analytical skills

→ Experts in selected analytical skills are among the winners.



NRA tasks. Sub-sample: Top fifth of the population distribution of skill $s_{i,\tau}$. Log wage change after occupational choices adjust.

3 **AI raises the value of social skills and lowers that of some analytical skills.**

Jobs exposed to AI shift toward collaboration and coordination.

Conclusion

A framework to quantify the labor market effects of AI

Literature

1 **On average, moderate AI exposure raises wages, high exposure lowers them.**

Task-level automation can be worker-level augmentation.

2 **Some workers face large losses, with reallocation offering limited insurance.**

Task automation transforms losers' usual alternative occupations in the same way.

3 **AI raises the value of social skills and lowers that of some analytical skills.**

Jobs exposed to AI shift toward collaboration and coordination.

Thank You!

Extra Slides

What's new? Position in the literature.

Back

We develop a framework to quantify job transformation effects and demonstrate their central role in shaping the labor market consequences of AI.

- **Job transformation** *[**Autor et al., 2003**; Spitz-Oener, 2006; Autor and Handel, 2013; Atalay et al., 2020; Autor et al., 2024; Gathmann et al., 2024]*
  - ⇒ **formalize & link with skills** → structurally quantify *wage effects*
- **Task-based theory** *[Acemoglu-Autor, 2011; Aghion-Jones-Jones, 2017; Acemoglu-Restrepo, 2018; **Acemoglu-Restrepo, 2022**; Hemous-Olsen, 2022; Freund, 2023; Hurst–Rubinstein–Shimizu, 2024; Autor-Thompson, 2025]*
  - ⇒ introduce jobs as task bundles → enable study of **job transformation**
- **Model-based analysis of AI** *[Acemoglu, 2025; **Hampole et al., 2025**; Fan, 2025; Restrepo, 2025; Althoff-Reichardt, 2025; Lashkari et al., 2026; Tonetti-Jones, 2026]*
  - ⇒ directly estimate granular skills → **demonstrate key role of JT**
- **Multi-dimensional skills** *[**Lindenlaub, 2017**; Lise-PostelVinay, 2021; Deming, 2023; Grigsby, 2023]*
  - ⇒ **estimate** distribution of task-specific skills → **tie skill returns to automation**

Job transformation: the case of weavers in the 19th century

Back

Period	Preparatory tasks		Tasks while machine running							Tasks while power loom stopped							
	Prepare warp	Dress warp	Let off warp	Pick shuttle	Beat reed	Take up cloth	Adjust warp tension	Replace empty bobbin	Monitoring	Fix smashes	Adjust temples	Back up loom	Replace empty shuttle	Fix broken weft	Fix broken warp end	Remove cloth, cleaning	Replace warp
Handloom	●	●	●	●	●	●	●		●		●		●	●	●	●	●
Early power loom (∼1820)							●	●	●	●	●	●	●	●	●	●	●
1833							●	●	●	●		●	●	●	●	●	●
1883							○	●	●	●			●	●	●	●	○

Notes. ● = Task performed; ○ = Reduced frequency; Empty = Task not performed.
Based on Bessen (2012)'s analysis of records of the Lawrence Company, MA.

Firm's optimal production problem

Back

- **Output** of firm in occ o with worker i given idiosyncratic shock $\varepsilon_{i,t} \sim \mathcal{N}(0, \varrho)$:

$$y_{i,o,t}(\cdot) = \underbrace{\prod_{\tau \in \mathcal{T}_l} \left(\exp\left(s_{i,\tau} + \varepsilon_{i,t}\right) \cdot \ell_{i,\tau,t}\right)^{\alpha_{o,\tau}}}_{\text{worker-produced}} \underbrace{\prod_{\tau \in \mathcal{T}_m} \left(\exp\left(z_\tau\right) \cdot m_{i,\tau,t}\right)^{\alpha_{o,\tau}}}_{\text{machine-produced}}$$

- **Profits:**

$$\pi_{i,o,t} = \max_{\{m_{i,\tau}\}_{\tau \in \mathcal{T}_m}, \{\ell_{i,\tau}\}_{\tau \in \mathcal{T}_l}} p_{o,t} y_{i,o,t}\left(\{\ell_{i,\tau,t}\}_{\tau \in \mathcal{T}_l}, \{m_{i,\tau,t}\}_{\tau \in \mathcal{T}_m}\right) - \exp\left(w_{i,o,t}\right) - r \sum_{\tau \in \mathcal{T}_m} m_{i,\tau,t}$$

$$\text{s.t.} \sum_{\tau \in \mathcal{T}_l} \ell_{i,\tau,t} = 1$$

Firm's optimal production problem

Back

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$$\text{s.t.} \sum_{\tau \in \mathcal{T}_l} \ell_{i,\tau,t} = 1$$

- **Optimality:**

FOC capital

$$\ell_{i,\tau,t} = \frac{\alpha_{o,\tau}}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}}$$

Firm's optimal production problem

Back

- **Output** of firm in occ o with worker i given idiosyncratic shock $\varepsilon_{i,t} \sim \mathcal{N}(0, \varrho)$:

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- **Optimality:**

FOC capital

$$\ell_{i,\tau,t} = \frac{\alpha_{o,\tau}}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}} \longrightarrow \textit{matrix A: } |\mathcal{O}| \times |\mathcal{T}_l|$$

Capital FOC and production

Back

- FOC for machines $M_{i,o,t} := \sum_{\tau \in \mathcal{T}_m} M_{i,o,\tau,t}$:

$$\left(\sum_{\tau \in \mathcal{T}_m} \alpha_{o,\tau} \right) \frac{p_{o,t} Y_{i,o,t}}{r} = M_{i,o,t}$$

and

$$M_{i,o,\tau,t} = \frac{\alpha_{o,\tau}}{\sum_{\tau \in \mathcal{T}_m} \alpha_{o,\tau}} M_{i,o,t}$$

- Plugging both FOCs into the production function yields

$$y_{i,o,t} = \log Y_{i,o,t} = \left[\sum_{\tau \in \mathcal{T}_l} \frac{\alpha_{o,\tau}}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}} s_{i,\tau} \right] + \varepsilon_{i,o} + \frac{1}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}} \log P_{o,t}$$
$$+ \left[\sum_{\tau \in \mathcal{T}} \frac{\alpha_{o,\tau}}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}} \log(\alpha_{o,\tau}) \right] - \log \left(\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau} \right) + \left[\sum_{\tau \in \mathcal{T}_m} \frac{\alpha_{o,\tau}}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}} (z_\tau - \log r) \right]$$

Equilibrium

Back

Remark: Equilibrium

An equilibrium is defined as a vector of prices $\vec{p}$ and a joint distribution G of occupation choices, log wages w, log skills s and idiosyncratic productivity shocks $\varepsilon_\cdot$, such that:

1. all firms make zero profits, i.e., at any point in the distribution:

$$w_{i,o,t} = \mu_o + \sum_{\mathcal{T}_l} \frac{\alpha_{o,\tau}}{LS_o} \cdot s_{i,\tau} + \varepsilon_{i,t}, \qquad Y \equiv \left(\sum_{o \in \mathcal{O}} \omega_o^{\frac{1}{\sigma}} Y_o^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} = \sum_{o \in \mathcal{O}} p_o Y_o$$

2. workers optimize, i.e., the marginal distribution of occupations conditional on wages follows

$$P(\hat{o} = o | w_{i,\cdot}) = \frac{\exp(w_{i,o}/\nu)}{\sum_{o'} \exp(w_{i,o'}/\nu)}$$

3. the unconditional marginal distributions of skills s and occupational shocks ε follow $\mathcal{N}(\bar{s}, \Sigma_s)$ and $\mathcal{N}(0, \varsigma^2 I)$, respectively.

Wage equation: details

Back

- Intercept

  $$\mu_o = \sum_{\tau \in \mathcal{T}} \frac{\alpha_{o,\tau}}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}} \log(\alpha_{o,\tau}) + \left(\sum_{\tau \in \mathcal{T}_m} \frac{\alpha_{o,\tau}}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}} (z_\tau - \log r) \right) + \frac{1}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}} \log P_{o,t}$$

  where $P_{o,t}$ is the price of output of occupation o

- We assume that in the initial steady state there is only one composite machine task with productivity normalized to $\log r$

Partial Automation

Back

- Our framework nests the concept of **partial automation**
  $\rightarrow$ only a fraction $\zeta_{\tau^\star} \in [0, 1]$ of task $\tau^\star$ can be automated
- **Modeling:** Equivalently, rewrite pre-automation production technology as

$$Y_{i,o,t} = \Gamma \cdot \prod_{\tau \in \mathcal{T}_m} \left(X^{\text{machines}}_{i,\tau,t}\right)^{\alpha_{o,\tau}} \cdot \prod_{\tau \in \mathcal{T}_\ell \setminus \{\tau^\star\}} \left(X^{\text{labor}}_{i,\tau,t}\right)^{\alpha_{o,\tau}} \cdot \left((1-\zeta_{\tau^\star})X^{\text{labor}}_{i,\tau^\star,t}\right)^{(1-\zeta_{\tau^\star})\alpha_{o,\tau^\star}} \cdot \left(\zeta_{\tau^\star} X^{\text{labor}}_{i,\tau^\star,t}\right)^{\zeta_{\tau^\star}\alpha_{o,\tau^\star}}.$$

  where $\Gamma = (1-\zeta_{\tau^\star})^{-\alpha_{o,\tau^\star}(1-\zeta_{\tau^\star})} \zeta_{\tau^\star}^{-\alpha_{o,\tau^\star}\zeta_{\tau^\star}}$
- Post-automation production technology becomes

$$Y_{i,o,t} = \Gamma \cdot \prod_{\tau \in \mathcal{T}_\ell \setminus \{\tau^\star\}} \left(X^{\text{labor}}_{i,\tau,t}\right)^{\alpha_{o,\tau}} \cdot \left(X^{\text{labor}}_{i,\tau^\star,t}\right)^{(1-\zeta_{\tau^\star})\alpha_{o,\tau^\star}} \cdot \left(X^{\text{machines}}_{i,\tau^\star,t}\right)^{\zeta_{\tau^\star}\alpha_{o,\tau^\star}} \cdot \prod_{\tau \in \mathcal{T}_m} \left(X^{\text{machines}}_{i,\tau,t}\right)^{\alpha_{o,\tau}}.$$

Extension to CES (work-in-progress)

Back

- Baseline assumes **Cobb-Douglas**
  - isolates the job transformation $\times$ skill specialization channel
  - time shares independent of skills $\rightarrow$ A measured directly from time use
- **CES across tasks** with elasticity ρ (Cobb-Douglas: $\rho \to 1$):

$$Y_{i,o,t} = \left(\sum_{\tau \in \mathcal{T}} \alpha_{o,\tau} X_{i,\tau,t}^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}}$$

- Time shares depend on skills ($\rho < 1$: more time on weak tasks)

$$\ell_{i,\tau,t} = \frac{\alpha_{o,\tau}^{\rho} \exp\left((\rho-1)s_{i,\tau}\right)}{\sum_{\tau' \in \mathcal{T}_l} \alpha_{o,\tau'}^{\rho} \exp\left((\rho-1)s_{i,\tau'}\right)} \overset{\rho=1}{=} \frac{\alpha_{o,\tau}}{\mathrm{LS}_o}$$

- **Wage equation:**

$$w_{i,o,t} = \log P_{o,t} + \frac{1}{\rho-1} \log \left(\frac{\sum_{\tau \in \mathcal{T}_l} \alpha_{o,\tau}^{\rho} \exp\left((\rho-1)s_{i,\tau}\right)}{\mathrm{LS}_{o,t}} \right) + \varepsilon_{i,t}$$

- **Our estimation approach can be extended to handle CES**
  - proposal from the linearized wage equation, importance weights correct for linearization

Extension to CES (work-in-progress): implications

- With complementary tasks ($\rho < 1$), output is limited (more strongly) by the least productive inputs
- **Implication 1: automating weak tasks benefits more, losing strong tasks hurts less**
  - baseline: automating tasks a worker is less skilled at raises their wage
  - CES reinforces this: the worker spent more time on this task
  - conversely, output is less sensitive to a worker's strong tasks, so automating them lowers the wage by less than under CD
- **Implication 2: gains from better machines are capped** (given P_o)
  - example: full automation of $\tau^\star$
  - as $z_{\tau^\star} \to \infty$, $\text{LS}'_o \to \text{LS}_o$: labor share recovers
  - but the remaining human tasks hold output back, so the wage gain from automating $\tau^\star$ is limited: $\Delta w_{i,o} \to -\log(1 - \ell_{i,\tau^\star})/(1-\rho)$
  - CD: LS falls and log wage rises by $\alpha_{o,\tau^\star}/\text{LS}'_o$ per unit of $z_{\tau^\star}$, without bound

Identification strategy

Back

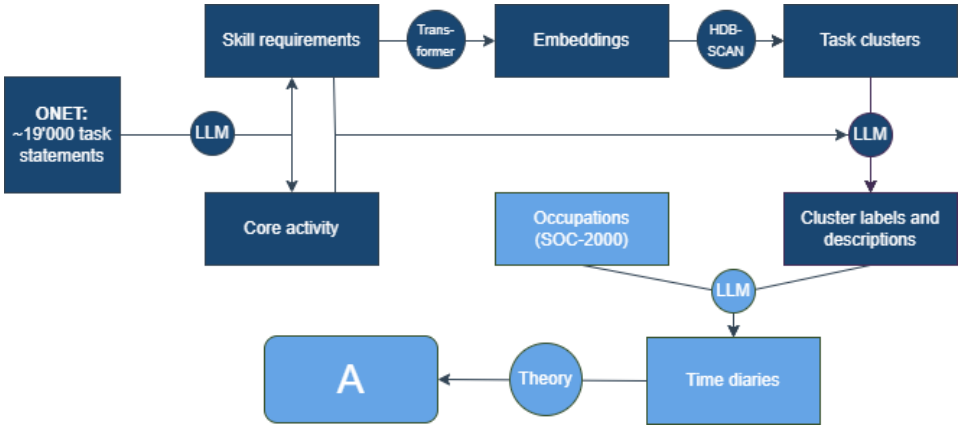
$$w_{i,o,t} = \mu_{o,t} + A_o s_i + \varepsilon_{i,t}, \quad s_i \sim \mathcal{N}(\bar{s}, \Sigma_s)$$

Parameter	Identifying variation	Key assumption
Σ_s	Same worker's wages in two years: **Stayers**: $\mathrm{Cov}(w_{i,o,t}, w_{i,o,t'} \mid \text{stay}) = A_o \tilde{\Sigma}_s A_o^\top$ $\rightarrow$ identifies *variance* of each occupation's skill composite $A_o s_i$ **Switchers**: $\mathrm{Cov}(w_{i,o,t}, w_{i,o',t'} \mid \text{switch}) = A_o \tilde{\Sigma}_s A_{o'}^\top$ $\rightarrow$ identifies *covariance* across occupations	A has full column rank; normal skills
ς	Wage growth of stayers: $\mathrm{Var}(\Delta w_{i,o,t} \mid \text{stay in } o) = 2\varsigma^2$	$\varepsilon_{i,t}$ common to all occ's
ν	Choice odds vs. the wage gap revealed in other years: $\ln\left[\Pr(o' \mid s_i) / \Pr(o \mid s_i)\right] = (w_{i,o',t} - w_{i,o,t})/\nu$	Logit taste shocks
$\bar{s}$	Change in mean wages: $\Delta \bar{w}_o^* = \Delta A_o \, \bar{s} + \text{const.}$	No other occ.-level shifts in 2000

$\tilde{\Sigma}_s$: skill covariance among the selected workers, mapped to population Σ_s through logit choice and normal skills.
* Adjusted for occ. selection, net of price change and known terms. $\bar{s}$ is identified up to its level (normalization).

Summary of step 1 pipeline

Back




Performing Skilled Manual Oper...
Operating, Calibrating, and In...
Developing and Delivering Inst...
Designing and Implementing Sys...
Preparing and Planning Meals
Providing Customer Service
Performing Clinical Procedures
Monitoring and Inspecting Syst...
Analyzing Natural Phenomena
Mediating and Consulting Clien...
Coordinating Administrative Ta...
Woodworkers
Other Construction and Related...
Entertainers and Performers, S...
Mathematical Science Occupatio...
Other Food Preparation and Ser...
Retail Sales Workers
Other Healthcare Practitioner...
Health Diagnosing and Treating...
Supervisors, Production Worker...
Media and Communication Equipm...
Supervisors, Personal Care and...
Supervisors, Construction and ...
Transportation, Tourism, and I...
Other Healthcare Support Occup...
0.6
0.5
0.4
0.3
0.2
0.1
0.0

Validation of LLM-generated time shares: overview

Back

1. LLM-generated task weights for (o, τ) highly correlated with the average importance rating that O*NET assigns to detailed tasks within each cluster ✓
2. Comparison of time share measurement: LLM vs BIBB survey ✓
3. Comparison of LLM-generated time shares for GWAs to O*NET importance weights ✓
4. Internal consistency: do measurements for detailed occupations aggregate up? ✓

Alternative A matrix: O*NET bottom-up

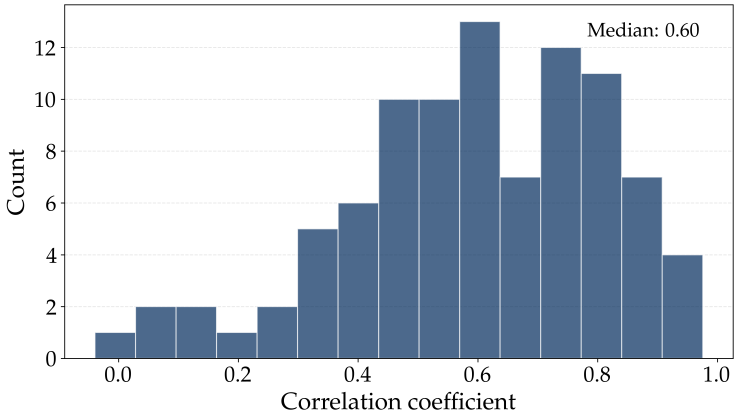
Back

- ONET contains (Likert scale) “task ratings” indicating the importance of detailed tasks *i* for *o*
- Create occupational task-weight matrices A_p (for $p \in \{2000s, 2010s, 2020s\}$, where each cell (o, τ) contains the average importance of task τ for occupation o
  - Notation
    - o' = detailed ONET occupation (8-digit), i = detailed ONET task, τ = cluster (model task)
    - $v \in V_p$ = ONET versions in period p
    - $r_{o',i,v} \in 1, 2, 3, 4, 5$ = raw importance rating of task i for occupation o' in version v; normalized to [0, 1]: $\tilde{r}_{o',i,v} = \frac{r_{o',i,v}-1}{4}$
    - $\mathcal{D}_{o,\tau,v}$ = set of (o', i) pairs in version v where o' maps to o and i maps to τ
  - Steps
    1. Sum weights across all (detailed occupation, detailed task) pairs: $S_{o,\tau,v} = \sum_{(o',i)\in\mathcal{D}_{o,\tau,v}} \tilde{r}_{o',i,v}$
    2. Average across versions within the period: $S_{o,\tau} = \frac{1}{|V_p|} \sum_{v\in V_p} S_{o,\tau,v}$
    3. Normalize to task shares within each occupation:

$$A_p(o, \tau) = \frac{S_{o,\tau}}{\sum_{\tau'} S_{o,\tau'}}$$

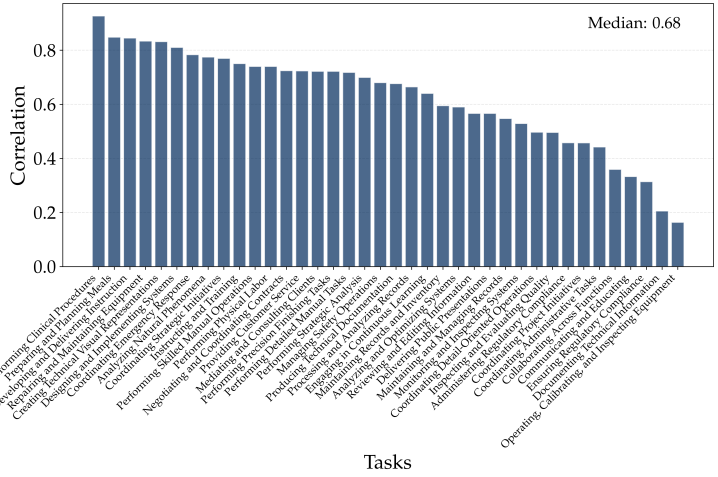
Validation: LLM task weights vs. ONET bottom-up – occupation level

Back



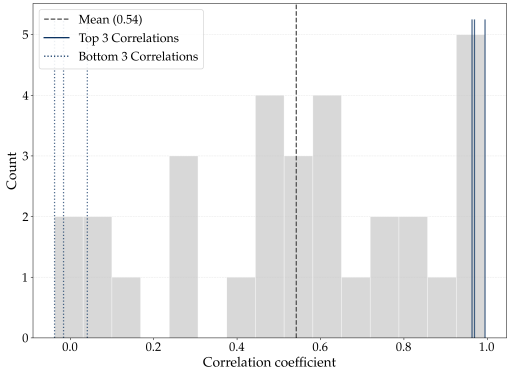
Validation: LLM task weights vs. ONET bottom-up – task-level

Back

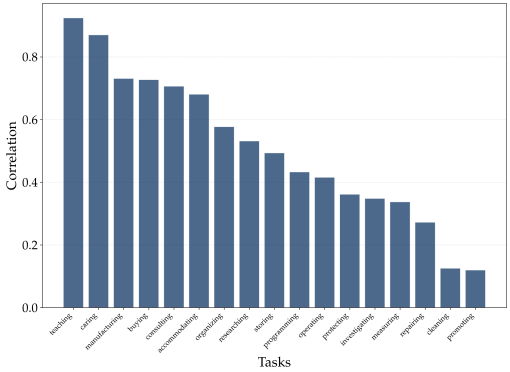


Validation: LLM-generated task shares vs. BIBB

Back



(a) Occupation-level correlations

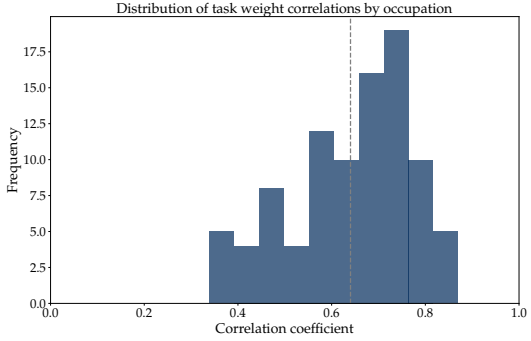


(b) Task-level correlations

Validation: O*NET GWAs (1)

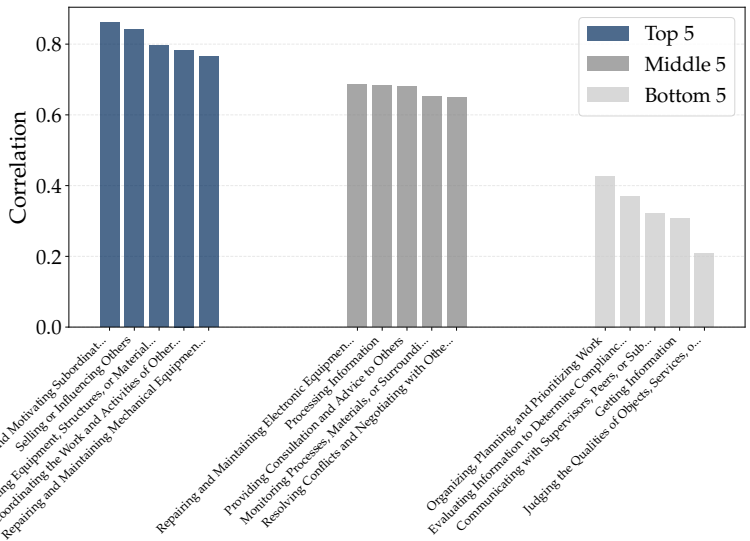
Back

- Take O*NET GWAs (O*NET 5.0, consistent with SOC-2000), construct relative importance for each GWA by occupation, aggregate to SOC-2000-3d
- Let LLM generate *time shares* for the GWAs for each SOC-2000-3d occ
- How do LLM-time shares correlate with vector of O*NET importance weights?



Validation: O*NET GWAs (2): correlation across occupations by task

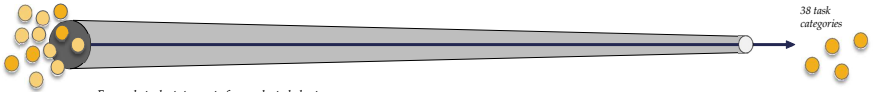
Back



Examples of mapping from detailed tasks to clusters

Back

We cluster O*NET's unstructured, detailed task statements into task categories based on similarity of skill requirements



Example task statements from selected clusters

Example task categories

- Smooth rough spots on walls and ceilings, using sandpaper.
- Lubricate moving parts on gate-crossing mechanisms and swinging signals.
- *Etc.*

Performing Detailed Manual Tasks

- Prepare reports of activities, evaluations, recommendations, or decisions.
- Prepare, examine, or analyze accounting records, financial statements, or other financial report.
- *Etc.*

Processing and Analyzing Records

- Confer with officials of public health and law enforcement agencies to coordinate interdepartmental activities.
- Confer with directors and production staff to discuss issues, such as production and casting problems, budgets, policies, and news coverage.
- *Etc.*

Coordinating Project Initiatives

For each task, we extract skill requirements, create semantic vector embeddings for these requirements using a transformer, and perform HDBSCAN-clustering (with hyperparameters tuned to maximize the DBCV score) on these embeddings to create broad task categories.

Task clustering: example tasks, extraction, assignment

Task	Activity	Skills	Cluster
Smooth rough spots on walls and ceilings, using sandpaper	smooth surfaces	manual dexterity (basic), attention to detail (basic)	Performing Detailed Manual Tasks
Lubricate moving parts on gate-crossing mechanisms and swinging signals	lubricate moving parts	manual dexterity (basic), attention to detail (basic)	Performing Detailed Manual Tasks
Perform physically demanding tasks, such as digging trenches to lay conduit or moving or lifting heavy objects	perform physical labor	physical endurance (advanced), manual dexterity (intermediate)	Performing Physical Labor
Prepare reports of activities, evaluations, recommendations, or decisions	prepare reports	report writing (advanced), analytical reasoning (intermediate), attention to detail (intermediate)	Processing and Analyzing Records
Confer with officials of public health and law enforcement agencies to coordinate interdepartmental activities.	coordinate interdepartmental activities	collaboration (advanced), project management (advanced), communication skills (intermediate)	Coordinating Project Initiatives

Details on the estimation strategy I

Back

- We consider two separate A-regimes (and occupational price vectors $(\vec{p}, \vec{p}')$)
- We assume that regimes change in the year $t^* = 2000$
- Exact likelihood:

$$\prod_i \int_s \left[\left(\int_{w_{i,\cdot,-\omega_\cdot}} \prod_t P(\hat{o}_{i,t} = \omega_{i,t} | w_{i,\cdot,\cdot,\nu}) \cdot f(w_{i,t,-\omega_t} | s, w_{i,\cdot,\omega_\cdot}, \varsigma, \vec{p}, \vec{p}') \right) \cdot f(s | w_{i,\cdot,\omega_\cdot}, \varsigma, \bar{s}, \Sigma_s, \vec{p}, \vec{p}') \right] \cdot f(w_{i,\cdot,\omega_\cdot} | \varsigma, \bar{s}, \Sigma_s, \vec{p}, \vec{p}')$$

- We maximize this likelihood while imposing that prices satisfy

$$H(\vec{p}, \vec{p}') = \Delta \log p_o + \frac{1}{\sigma} \Big(\Delta \log Y_o(\vec{p}, \vec{p}') - \Delta \log Y(\vec{p}, \vec{p}') \Big) = 0$$

Details on the estimation strategy II

Back

- **Strategy:** Monte Carlo integration - for all i generate n_o draws from

$$f(w_{i,\cdot,-\omega.}|w_{i,\cdot,\omega.},\varsigma,\bar{s},\Sigma_s,\vec{p},\vec{p}') = \int_s f(w_{i,\cdot,-\omega.}|s,w_{i,\cdot,\omega.},\varsigma,\vec{p},\vec{p}')f(s|w_{i,\cdot,\omega.},\varsigma,\bar{s},\Sigma_s,\vec{p},\vec{p}')$$

and evaluate the mean of $P(\hat{o}_{i,t} = \omega_{i,t}|w_{i,\cdot,t},\nu)$ to obtain an estimator for $\mathcal{L}_i(\theta)$:

$$\hat{\mathcal{L}}_i(w_{i,t,\omega},\nu,\varsigma,\bar{s},\Sigma_s,\vec{p},\vec{p}') = \left(\frac{1}{n_o}\sum_j\prod_t P(\hat{o}_{i,t} = \omega_{i,t}|w_{j,t,\cdot},\nu)\right)\cdot f(w_{i,\cdot,\omega.}|\varsigma,\bar{s},\Sigma_s,\vec{p},\vec{p}')$$

- We use the implicit function theorem to compute an augmented gradient that accounts for the GE restriction:

$$\bar{D}_x\mathcal{L} = D_x\mathcal{L} + D_{\vec{p}'}\mathcal{L}\cdot(-D_{\vec{p}'}H)^{-1}\cdot D_xH \quad \forall x \in \{\nu,\varsigma,\bar{s},\Sigma_s,\vec{p}\}$$

Details on the estimation strategy III

Back

- Two numerical techniques help speed up the maximum likelihood computation
- **Auto-differentiation:** efficiently compute the gradient of this function
- **Stochastic gradient descent:**
  - basic technique: gradient descent

$$\theta_{t+1} = \theta_t - \eta \cdot \nabla \left(-\mathcal{L}(\theta_t) \right)$$

  - randomly partition individuals into n groups:

$$\{1, 2, \ldots, I\} = B_1 \cup B_2 \cup \ldots \cup B_n, \quad B_i \cap B_j = \emptyset$$

  - calculate the likelihood based on batch $B_1, \ldots, B_n$ only
  - when done, draw a new partition

Brief summary of NLSY '79 data

Back

- 6,033 workers, 1979-2018, 110,618 total observations
  - drop individuals in the military sample and the minority oversample
- Construct annual panel comprising each individual's primary job (if any)
  - we focus on full time jobs only
- Harmonized occupational classification at SOC-2000 "minor groups" level (# 93)
  - we drop very small occupations ($<$¡0.3% employment share)
- Average st. dev. of log wages: 0.60
- Average annual moving probability $\sim$ 37%

Parameter values

Back

- We calibrate the cross-occupation EoS $\sigma = 2$ following Burstein-Morales-Vogel (2019) who estimate $\sigma \in \{1.81, 2.10\}$.
- For the scalar parameters, we estimate $\nu = 0.20$ and $\varrho = 0.30$.
- $\nu = 0.20$ implies that reducing prospective wages in a given occupation by 1% lowers the odds of choosing this occupation by about 5%
- $\varrho = 0.30$ indicates that a one-standard-deviation occupation-specific random productivity shock can raise or lower wages by about 30% in a given year.

Task categorization following Autor et al. (2003)

Back

- **Mapping to conventional task categories:** Following ALM2003 and others, let category $c \in \{NRA, NRI, NRM, RC, RM, U\}$, where "U" stands for "unclassified"
  - We use an LLM pipeline to map each model task τ to categories c.

- **Individual-level task shares:** Define the task share of task τ for worker i in year t as $\tilde{\alpha}_{o(i,t),\tau} = \frac{\alpha_{o(i,t),\tau}}{\sum_{\tau \in \mathcal{T}_l} \alpha_{o(i,t),\tau}}$, where $o(i,t)$ is the occupation chosen by i in year τ. Define individual-year specific task indices that measure the "intensity" of category c in worker i's job in year t, consistent with to surveys *[Spitz-Oener, 2006; Antonczyk et al., 2006]*
  - Taskindex-unweighted$_{ict} = \frac{\sum_{\tau \in \mathcal{T}} \mathbf{1}\{\tau \in c\} \cdot \mathbf{1}\{\tilde{\alpha}_{o(i,t),\tau} \geq \text{threshold}\}}{\sum_{\tau \in \mathcal{T}} \mathbf{1}\{\tilde{\alpha}_{o(i,t),\tau} \geq \text{threshold}\}}$, where "threshold" is a minimum time-share requirement (default: 0.1).
  - Taskindex-weighted$_{ict} = \frac{\sum_{\tau \in \mathcal{T}} \mathbf{1}\{\tau \in c\} \cdot \tilde{\alpha}_{o(i,t),\tau}}{\sum_{\tau \in \mathcal{T}} \tilde{\alpha}_{o(i,t),\tau}}$.

Skill indices and regressions

Back

- **Construction of skill measures:** For each worker *i*, we construct aggregate skill measures in three steps, aligned with literature *[Deming, 2017; Edin et al., 2022]*
  1. Standardize each task-specific log skill $s_{i\tau}$ in the population: $\tilde{s}_{i\tau} = \frac{s_{i\tau} - \bar{s}_\tau}{\sigma_{s_\tau}}$
  2. Average standardized skills within category *c*: $\bar{s}_{ic}^{\text{raw}} = \frac{1}{|\mathcal{T}_c|} \sum_{\tau \in \mathcal{T}_c} \tilde{s}_{i\tau}$
  3. Re-standardize the aggregate: $S_{ic} = \frac{\bar{s}_{ic}^{\text{raw}} - \overline{\bar{s}_c^{\text{raw}}}}{\sigma_{\bar{s}_c^{\text{raw}}}}$

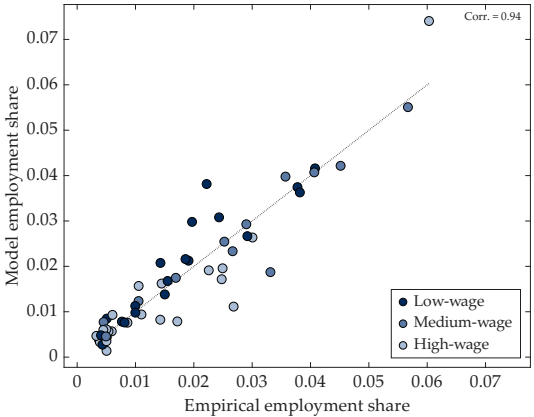
- **Wage regression:** straightforward pooled OLS wage regression

$$\log w_{it} = \delta_t + \sum_{c \in \mathcal{C}} \beta_c S_{ic} + \varepsilon_{it} \tag{1}$$

- **Interpretation:** Coefficients represent the change in log wage associated with a one standard deviation increase in the respective skill measure

Occupational employment shares

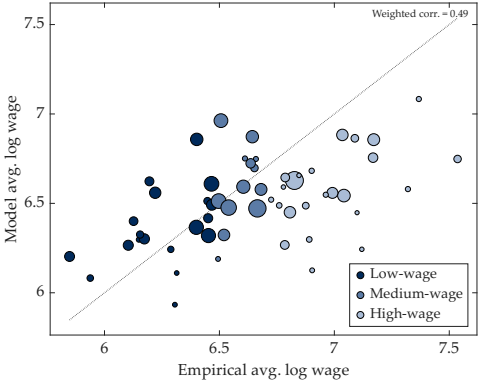
Back



Wages: dispersion & occ. level averages

Back

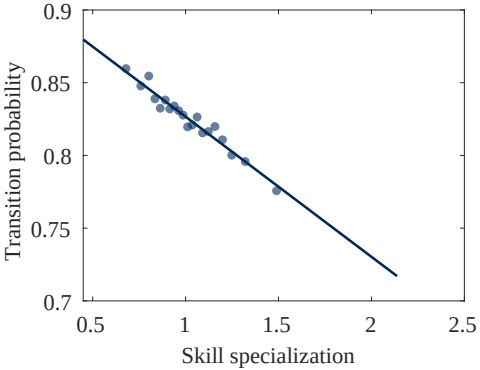
- Wage variance decomposition: model moments reasonably aligned with data
  - data: std. dev. 0.54, 30% between-occ. share
  - model: std. dev 0.57, 13% between-occ. share



Model properties: specialization shapes switching frequency

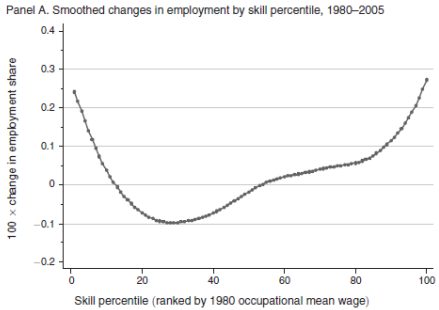
Back

→ Skill specialization tends to generate persistence in occupational choice *[Kambourov and Manovskii, 2008; Geel et al., 2011]*



RBTC effects in the literature: employment polarization

→ The empirical literature documents a U-shaped change in employment by "skill" (wage percentile) & attributes it to RBTC



(a) Data *[Autor & Dorn, 2013]*



FIGURE 1. PERCENTAGE CHANGES IN EMPLOYMENT SHARES OVER 1993–2006 FOR JOBS RANKED BY THEIR 1994 LOG WAGE

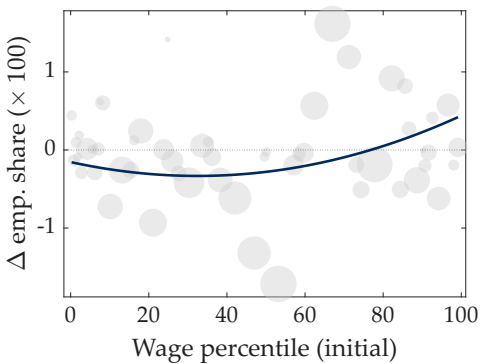
Note: Jobs are industry-occupation cells weighted by their 1993 employment shares, pooled across countries, and ranked by their UK 1994 log mean wage.

Sources: European Union Labour Force Survey 1993–2006, United Kingdom Labour Force Survey 1994.

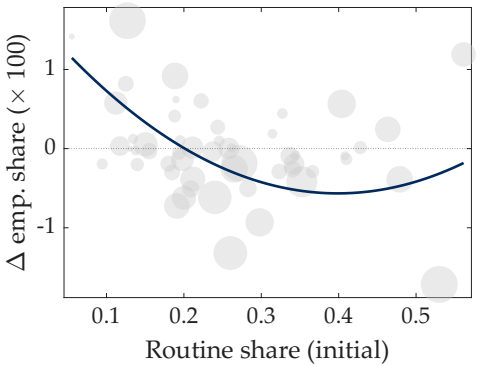
(b) Data *[Goos et al., 2009]*

Model matches empirical employment polarization

→ Model matches this U-shaped change in employment & decline in routine jobs



(a) Model: by wage percentile



(b) Model: by routine share

Historical validation: the case of RBTC – task changes

Back

- **Model analysis:** compare SS under post-2000 vs. pre-2000 *A*
- Model captures RBTC = validation of our *A* measures & model mechanisms

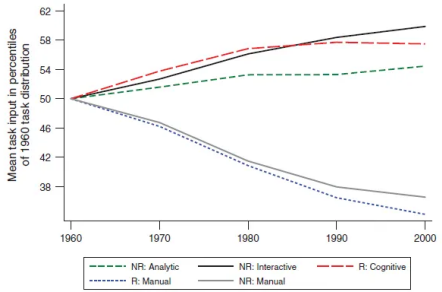
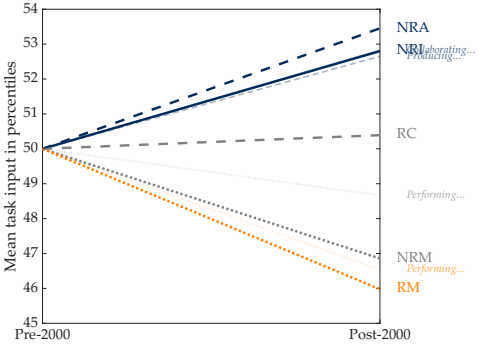


FIGURE 3. COMPARISON TO AUTOR, LEVY, AND MURNANE (2003, FIGURE 1)

(a) Data *[Atalay et al., 2020]*



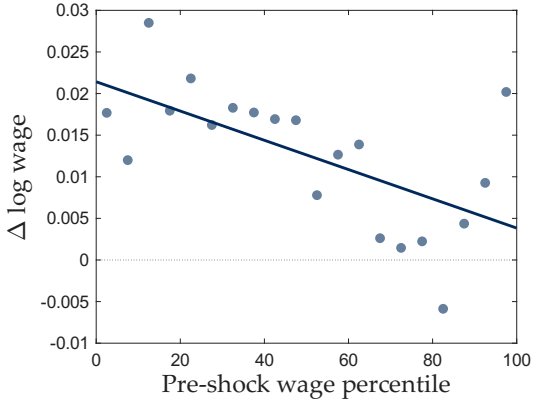
(b) Model

4 **Distribution.** Will high-wage or low-wage workers do better due to AI?

AI shock is mildly “progressive”

Back

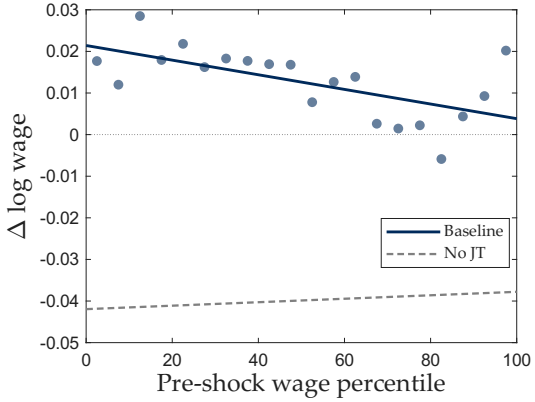
→ Wage gains disproportionately accrue to low-wage earners



AI shock is mildly "progressive" *because* of JT

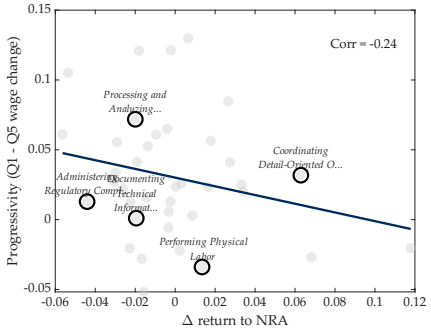
→ Wage gains disproportionately accrue to low-wage earners—once you account for JT

→ JT also generates increase in *average* wage = **productivity effect**

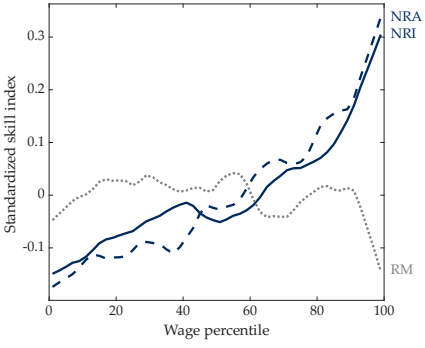


This reflects shifts in the returns to skills prevalent among high earners

- AI is (mildly) progressive
  - ...as it lowers the return to analytical skills, which are over-represented at the top
  - ...but raises the return to social skills, which follow a similar pattern



(a) Comparing shocks

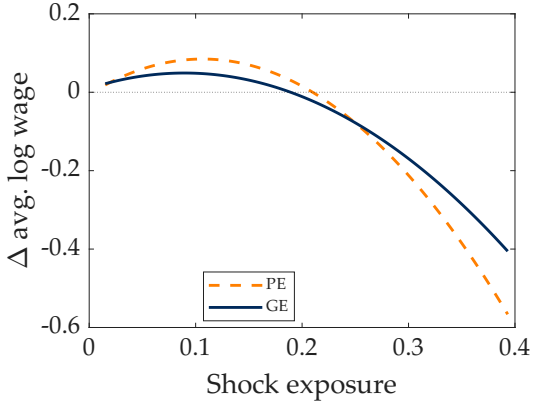


(b) Skills along the distribution

4 **Distribution.** The distributional impact of genAI is mildly progressive

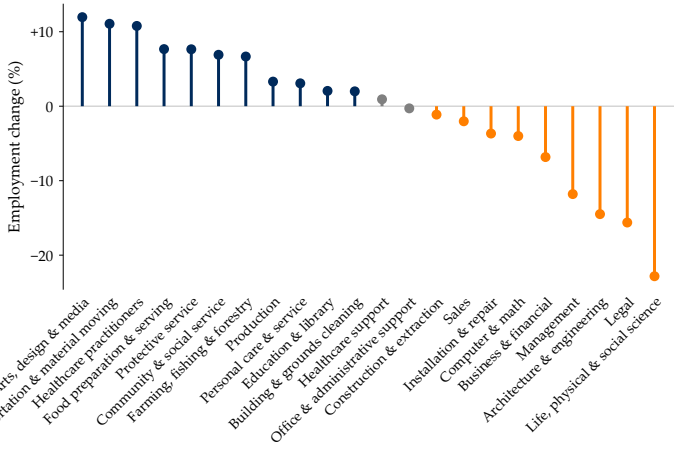
The role of GE

Back



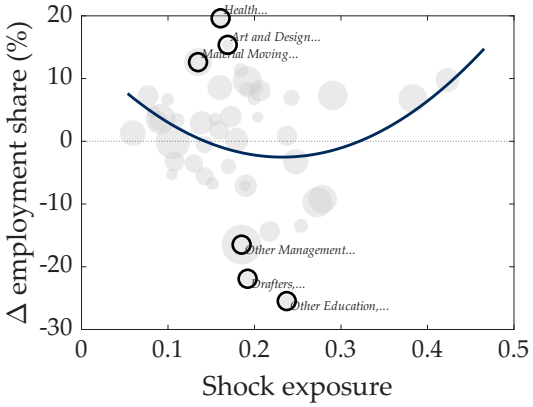
Employment falls most in white-collar professions

Back



Percent change in employment by two-digit occupation (SOC-2000 major groups), after vs. before the AI shock.

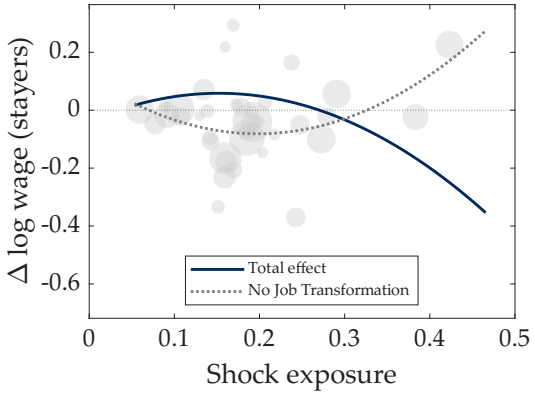
Employment effects



Job transformation drives this pattern

Back

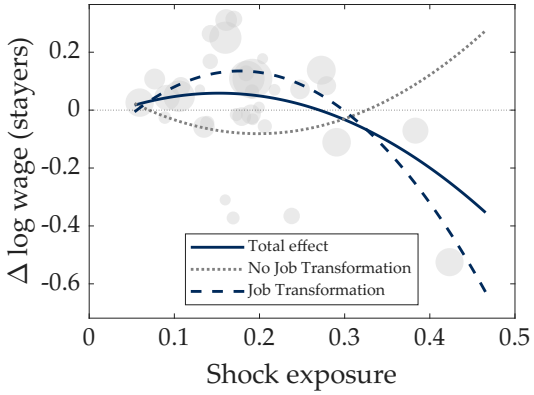
→ Holding *A* fixed, i.e. absent job transformation, the inverted-U shape disappears



Focusing on stayers for clarity of exposition.

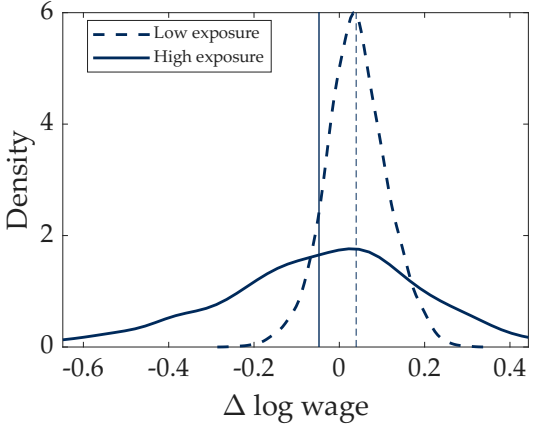
Job transformation interacts with selection

$\rightarrow$ Job transformation ($\Delta A \cdot s$) is *positive* for moderate exposure (AI complements what incumbents are best at) but *negative* at high exposure (AI takes over what incumbents are best at)



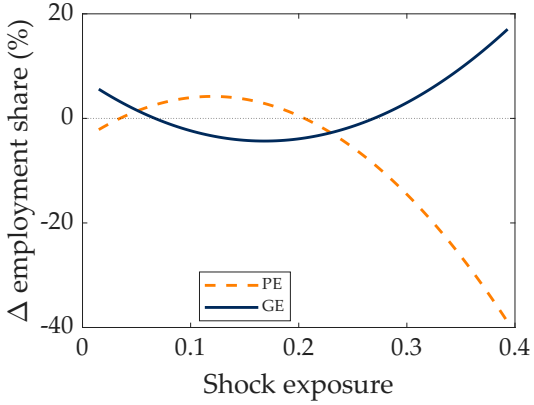
The average masks large heterogeneity

→ Dispersion in outcomes is greater among workers in high-exposure occupations



Employment: the role of GE

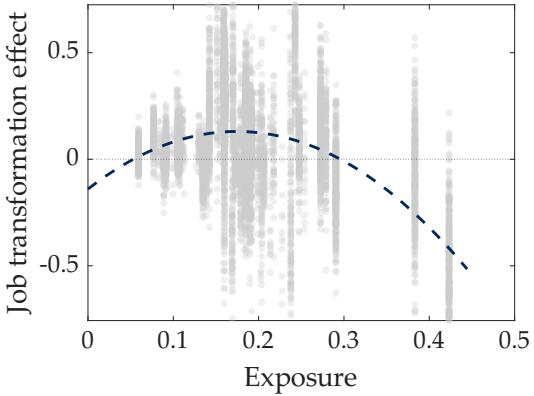
Back



Dispersion in outcomes arises from job transformation

Back

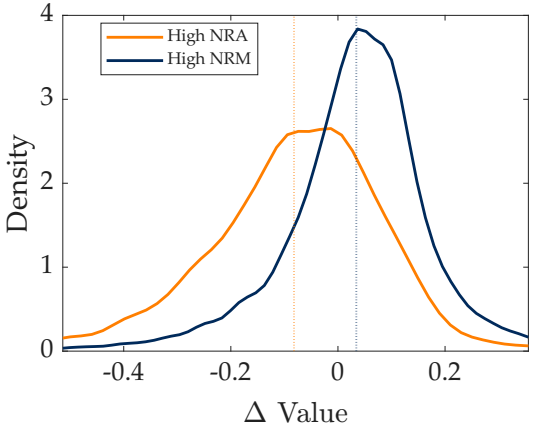
→ Dispersion conditional on exposure arises from job transformation as opposed to occupation-level shifts (which ∼ by definition don't generate individual-level dispersion)



These shifts help predict who the winners & losers tend to be

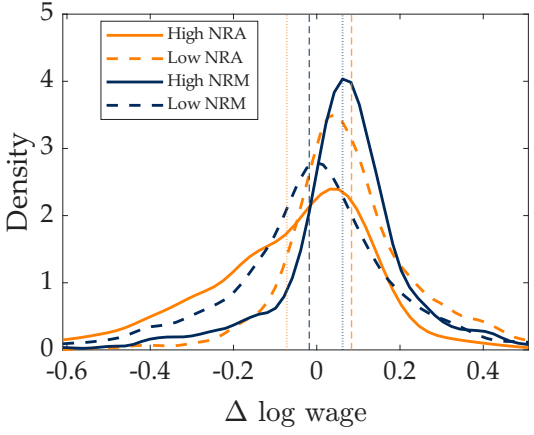
Wage version

→ High-NRA workers are over-represented among losers, high-NRM workers are over-represented among winners



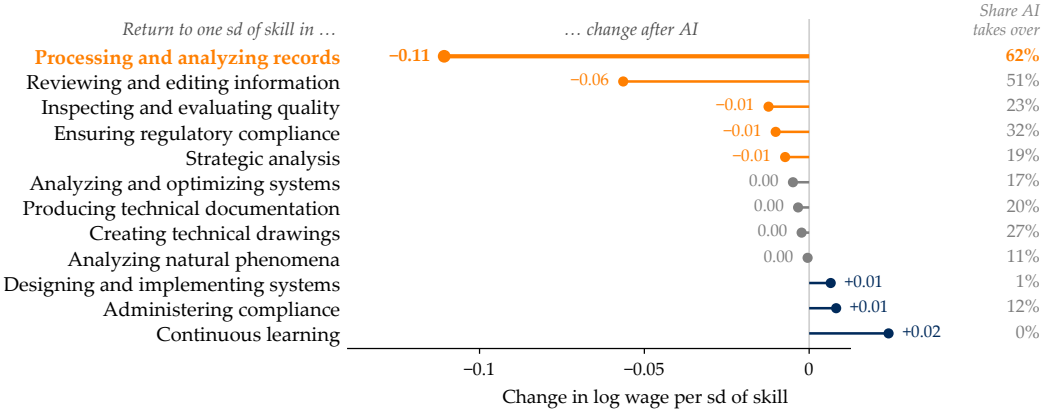
Shifts in returns explain who winners & losers are (wage version)

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Changes in skill returns for analytical tasks

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Return to skill τ: $\partial V_i / \partial s_{i,\tau} = \sum_o P_{i,o} A_{o,\tau}$, averaged over workers and scaled by the population standard deviation of $s_{i,\tau}$. Before AI: pre-shock choice probabilities $P_{i,o}$ and weights A. After AI: post-shock $P_{i,o}$ and A′.

Skilled Trades may be AI-proof

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"In many of the European countries, for example, if you look at how much an electrician can make versus someone that has a professional clerical job, which can be jeopardised by AI, the electrician can make more money." [...] " supports a view of the future where skilled trades and construction are resilient"

Future-proof your career and get a blue-collar job

Plus, what to do when a popular boss moves on



Fixing up a secure future? © FT montage? Dreamstime

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